simulated by Paul Franklin's Double Negative team using relativity equations I gave to them, is true. This is what it really would look like.

By contrast, the blight that endangers human life on Earth in *Interstellar* (Figure 3.4 and Chapter 11) is an educated guess in one sense, and a speculation in another. Let me explain.

Throughout recorded history, the crops that humans grow have been plagued by occasional blights (rapidly spreading diseases caused by microbes). The biology that underlies these blights is based on chemistry, which in turn is based on the quantum laws. Scientists do not yet know how to deduce, from the quantum laws, *all* of the relevant chemistry (but they can deduce *much* of it); and they do not yet know how to deduce from chemistry all of the relevant biology. Nevertheless, from observations and experiments, biologists have learned much about blights. The blights encountered by humans thus far have not jumped from infecting one type of plant to another with such speed as to endanger human life. But nothing we know guarantees this can't happen. That such a blight is possible is an *educated guess*. That it might someday occur is a *speculation* that most biologists regard as very unlikely.